Technical University of Munich

Exercises Rechnereinführung WS 2026/27

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3 – Markup Languages (LaTeX)

Exercises marked with a (*) are bonus tasks and should only be done if there is enough time left.

3.1 LaTeX

3.1.1 Overleaf

In the context of this exercise, we will compile LaTeX using Overleaf. You can either create an Overleaf account, or use TUM's own Overleaf/ShareLaTeX instance.

Create a new "Blank Project". Take a look at the contents of `main.tex` and familiarize yourself with Overleaf's UI.

First import the `babel` package, with which you can set the language of your document:

```
\usepackage[english]{babel}
```

Then try to recreate the following within the "Introduction" section:

TUM has multiple campus locations:

- München
- Garching
- Weihenstephan
- Straubing
- Heilbronn
- Singapur

Solution

```
TUM has multiple campus locations:
\begin{itemize}
    \item Munich
    \item Garching
    \item Weihenstephan
    \item Straubing
    \item Heilbronn
    \item Singapore
\end{itemize}
```

3.1.2 LaTeX Math

An incomplete list of mathematical symbols in LaTeX can be found here, while the KaTeX documentation can also be a good source. If you cannot find the correct symbol, you can also draw and find them with DeTeXify.

1. Create the following formulas in LaTeX. To verify your solution, use your `main.tex` from exercise 3.1.1.[1]

   Write your formula within *Display-Math* brackets (`\[ ... \]`). Also add the `amsmath` and `amsfonts` packages, which define several math constructs like matrices:

   ```
   \usepackage{amsmath}
   \usepackage{amsfonts}
   ```

   a)
   $$x_{1,2} = \frac{p}{2} \pm \sqrt{\frac{p^2}{4} - q}$$

   b)
   $$x_{1,2} = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

   c)
   $$\sum_{k=0}^{\infty} \frac{x^k}{k!} = e^x$$

   d)
   $$\sum_{i=0}^{\infty} (-1)^i \frac{x^{2i}}{(2i)!} = \sin(x)$$

[1] Alternatively you can type your formulas on the KaTeX Website to preview them.

e)

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!}(x-a)^n$$

f)

$$R\mathbf{v} = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix} \cdot \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x\cos\theta - y\sin\theta \\ x\sin\theta + y\cos\theta \end{pmatrix}$$

g)

$$\left(\begin{array}{cc|c} 1 & 2 & 3 \\ 4 & 5 & 6 \end{array}\right)$$

Note: use `\vrule`

2.

3. Use *Inline-Math* (`\(...\)`) to write the following propositions:
   a) Let $n \in \mathbb{N}_0$ be arbitrary but fixed. Then $n! = n \cdot (n-1)!$.
   b) Let x be a local minimum of f.
   Then: $\exists \epsilon > 0 : \forall x' \in \mathbb{B}_\epsilon(x) : f(x') \geq f(x)$.

4. Try further examples with TEXnique.

Solution

1. a)
```
x_{1, 2} = \frac{p}{2} \pm\sqrt{\frac{p^2}{4} - q}
```
   b)
```
x_{1, 2} = \frac{-b \pm\sqrt{b^2 - 4ac}}{2a}
```
   c)
```
\sum_{k=0}^{\infty} \frac{x^k}{k!} = e^x
```
   d)
```
\sum_{i=0}^{\infty} {(-1)}^i \frac{x^{2i}}{(2i)!} = \sin(x)
```
   e)
```
f(x) = \sum_{n=0}^\infty\frac{f^{(n)}(a)}{n!}{(x-a)}^n
```
   f)
```
R \mathbf{v} = \begin{pmatrix}
    \cos\theta & -\sin\theta \\
    \sin\theta & \cos\theta
\end{pmatrix}
\cdot \begin{pmatrix} x \\ y \end{pmatrix}
= \begin{pmatrix}
    x\cos\theta - y\sin\theta \\
    x\sin\theta + y\cos\theta
\end{pmatrix}
```

g)
```
\begin{pmatrix}
    1 & 2 & \vrule & 3 \\
    4 & 5 & \vrule & 6
\end{pmatrix}
```

2. a)
```
Let \(n \in \mathbb{N}_0\) be arbitrary but fixed.
Then \(n! = n \cdot (n-1)!\).
```

b)
```
Let \(x\) be a local minimum of \(f\).\\
Then: \(\exists \epsilon > 0:
\forall x' \in \mathbb{B}_\epsilon(x): f(x') \ge f(x)\).
```

3.1.3 References and Citation

To do citations in LaTeX, we usually use one of two systems:

- `bibtex`
- `biber`

Where `biber` is the more modern solution of the two. In both cases however, the `biblatex` package is used.

3.1.3.1 Creating a document with references

Proceed as follows:

1. Create a file `biblio.bib`.
2. Add different sources to this file. Use the ACM Digital Library, Google Scholar, dblp, IEEE Xplore and arXiv to export different papers and books directly into the Bibtex format (usually you can also get access to the papers themselves).

   E.g. try to use the Dragon Book as a source.
3. Write the following content into the `main.tex` file:

```
\documentclass{article}
\usepackage[utf8]{inputenc}
\usepackage[english]{babel}

\begin{document}

\end{document}
```

4. To do citations in LaTeX, you will need the `biblatex` package:

```
\usepackage[backend=biber]{biblatex}
```

   You can remove `backend=biber` to use `bibtex` instead of `biber`.

5. To be able to add citations to your chosen sources, you will need to tell LaTeX your `.bib` file (in this case `biblio.bib`, but you can name the file however you want):

```
\usepackage[backend=biber]{biblatex}
\addbibresource{biblio.bib}
```

6. To get LaTeX to print a list of your sources, you will need to add the `\printbibliography` command wherever you want it to be (typically right before `\end{document}`).

7. Now you can add the `\cite{<NAME_OF_SOURCE>}` command in your document.

Solution

Example Document:

```
% biblio.bib
@book{dragon-book,
    author = {Alfred V. Aho and Monica S. Lam and Ravi Sethi and Jeffrey D.
    ↪  Ullman},
    title = {Compilers: Principles, Techniques, and Tools},
    year = {2006},
    publisher = {Pearson Education},
    edition = {2},
    isbn = {978-0-321-48681-3},
}

@book{pierce-types,
    author = {Benjamin C. Pierce},
    title = {Types and Programming Languages},
    year = {2002},
    publisher = {The MIT Press},
    edition = {1},
    isbn = {978-0-262-16209-8},
}

@book{kernighan-ritchie,
    author = {Brian W. Kernighan and Dennis M. Ritchie},
    title = {The C Programming Language},
    year = {1988},
    publisher = {Prentice Hall},
    edition = {2},
    isbn = {978-0-13-110370-2},
}
```

```latex
% main.tex
\documentclass{article}
\usepackage[utf8]{inputenc}
\usepackage[english]{babel}
\usepackage[backend=biber]{biblatex}
\addbibresource{biblio.bib}

\begin{document}

Compiler design is a useful field to study, not just for building compilers,
↪  but other disciplines in Computer Science as
↪  well~\cite[Chapter~1.5]{dragon-book}.

\begin{quote}
    Parametric polymorphism [...] allows a single piece of code to be typed
    ↪  ``generically,'' using variables in place of actual types, and then
    ↪  instantiated with particular types as needed.
\end{quote}~\fullcite[Chapter~23.2]{pierce-types}

\printbibliography

\end{document}
```

3.2 Markdown

Use Zulip to exchange math formulas that you learned in school.

Here are a few examples:

- Perfectly elastic impact:

$$u_1 = \frac{m_1 \cdot v_1 + m_2 \cdot (2v_2 - v_1)}{m_1 + m_2}$$

- Newton's law of gravitation:

$$F_G = G \cdot \frac{m_1 \cdot m_2}{r^2}$$

- Wave equation of a linear standing wave:

$$y(t; x) = 2 \cdot A \cdot \cos\left(\frac{2\pi \cdot x}{\lambda}\right) \cdot \sin\left(\frac{2\pi}{T} \cdot t\right)$$

- Distribution function of the normal distribution:

$$F(x) = \frac{1}{\sigma\sqrt{2\pi}} \cdot \int_{-\infty}^{x} e^{-\frac{1}{2}\left(\frac{t-\mu}{\sigma}\right)^2} dt$$

Solution

On Zulip, there are two different ways to write math:

1. For inline math one can use $$ at the beginning and $$ at the end of the block. E.g. `$$a^2 + b^2 = c^2$$` (in LaTeX, $$ would be a *display*-block!)
2. For display math one can use a `math` code block. E.g.:

````
```math
a^2 + b^2 = c^2
```
````

If you click on the question mark in the bottom left of a Zulip chat, you can open the formatting help which has the same information.
The given examples would be solved like so:

- `$$u_1 = \frac{m_1 \cdot v_1 + m_2 \cdot (2v_2 - v_1)}{m_1 + m_2}$$`
- `$$F_G = G \cdot \frac{m_1 \cdot m_2}{r^2}$$`
- ````
  ```math
  y(t;x) = 2 \cdot A \cdot \cos \left( \frac{2\pi \cdot x}{\lambda}
  ↪  \right)
      \cdot \sin \left( \frac{2\pi}{T} \cdot t \right)
  ```
  ````
- ````
  ```math
  F(x) = \frac{1}{\sigma \sqrt{2\pi}} \cdot \int^x_{- \infty}
      e^{-\frac{1}{2} \left( \frac{t - \mu}{\sigma} \right)^2} dt
  ```
  ````

3.3 (*) LaTeX Locally

While we used LaTeX through Overleaf so far, it is possible to compile LaTeX documents locally within a terminal.

3.3.1 LaTeX Setup

1. Install a LaTeX distribution on your Computer. Which distribution you choose depends on your operating system:
   - Linux: Texlive (e.g. with your package manager)
   - MacOS: MacTeX
   - Windows: MiKTeX
2. Create a file `main.tex` with the following content:

```latex
\documentclass{article}
\usepackage[utf8]{inputenc}
\usepackage[english]{babel}

\begin{document}
    Hello \LaTeX{}!
\end{document}
```

3. Compile the file with `pdflatex`

Solution

To compile `main.tex` with `pdflatex`, is suffices to run `pdflatex main.tex`.

3.3.2 References and Citation

To compile LaTeX documents with references, it is not enough anymore to just call the compiler (like `pdflatex`). Instead, you need to compile in 4 steps:

1. `pdflatex`
2. `biber`
3. `pdflatex`
4. `pdflatex`

Since this is quite cumbersome to do after each change of your document, you can instead use `latexmk` which does this for your.

The setup for a LaTeX file with refernces is identical to that of exercise 3.1.3.

Solution

It's best to compile with `latexmk`. E.g. to compile `main.tex`: `latexmk -pdf main.tex`.